RNG Comparative Validation Report

BBRES-RNG | Java SecureRandom | Java Math.random()

45-Test Full Statistical Validation Suite

Report Date	April 05, 2026
BBRES-RNG Developer	Mehul Singh
Bit Sample Size (each)	10,918,505 bits
Integer Sample Size (each)	100,000 integers [0..999]
Total Tests per RNG	45 (135 total across all three)
Significance Level	α = 0.01 (99% Confidence)

1. Executive Summary

This report presents a comparative statistical analysis of three RNG architectures subjected to an identical battery of 45 tests: BBRES-RNG (Bit Based Randomized Entropy System), a scheduler-based entropy harvesting RNG developed by Mehul Singh; Java SecureRandom, the standard CSPRNG in the Java platform; and Java Math.random(), a Linear Congruential Generator (LCG) from the Java standard library. Each was tested on the same 45-test battery encompassing NIST SP 800-22 (core and extended), uniformity, spectral, entropy, adversarial ML, cryptographic wrapper, and integer-level analyses.

RNG	Tests Passed	Verdict	Rank	Rating	Crypto
BBRES-RNG	**45 / 45**	**ACCEPTED**	**#1**	★★★★★	**YES**
Java SecureRandom	**44 / 45**	**REVIEW NEEDED**	**#2**	★★★★☆	**YES**
Java Math.random()	**44 / 45**	**REVIEW NEEDED**	**#3**	★★★★☆	**NO**

BBRES-RNG is the only generator to achieve a perfect 45/45 pass rate. It is ranked first due to its complete pass across every test category, including the Spectral FFT (which SecureRandom failed) and the Anderson-Darling test (which Math.random() failed). Both SecureRandom and Math.random() recorded 44/45, each failing in different areas. BBRES-RNG is confirmed suitable for cryptographic use; Java SecureRandom is also suitable despite one spectral anomaly; Java Math.random() is not suitable for cryptographic applications due to its LCG architecture.

2. Category-by-Category Comparison

2.1 NIST SP 800-22 Core Tests

All three generators passed all nine NIST core tests. No anomalies were detected in frequency, runs, block frequency, longest run, cumulative sums, approximate entropy, or serial tests.

Test	BBRES-RNG	SecureRandom	Math.random()	Status
Frequency (Monobit)	0.441897	0.214121	0.814095	ALL PASS
Block Frequency (M=128)	0.143304	0.855332	0.710312	ALL PASS
Runs Test	0.670559	0.375794	0.521743	ALL PASS
Longest Run of Ones	0.046389	0.875194	0.926790	ALL PASS
Cumulative Sums (Fwd)	0.164091	0.227216	0.943917	ALL PASS
Cumulative Sums (Rev)	0.656919	0.259846	0.870897	ALL PASS
Approximate Entropy (m=2)	0.851301	0.648893	0.537255	ALL PASS
Serial (m=2) - delta1	0.679895	0.312568	0.792248	ALL PASS
Serial (m=2) - delta2	0.671131	0.376372	0.521732	ALL PASS

2.2 NIST SP 800-22 Extended Tests

All three generators passed all seven extended NIST tests.

Test	BBRES-RNG	SecureRandom	Math.random()	Status
Maurer's Universal Statistical	0.722786	0.185709	0.882058	ALL PASS
Poker Test (m=4)	0.517531	0.132294	0.676137	ALL PASS
Random Excursion Variant	0.933697	0.285790	0.376274	ALL PASS
Binary Matrix Rank	0.982609	0.421437	0.617351	ALL PASS
Non-Overlapping Template (m=9)	0.361738	0.183847	0.399037	ALL PASS
Overlapping Template (m=9)	1.000000	1.000000	1.000000	ALL PASS
Linear Complexity	0.620280	0.600333	0.329608	ALL PASS

2.3 Distribution and Uniformity Analysis

All three generators passed all uniformity tests at byte, nibble, and 2-bit granularities.

Test	BBRES-RNG	SecureRandom	Math.random()	Status
Byte-Level Chi-Square	0.324837	0.434295	0.863830	ALL PASS
Nibble-Level Chi-Square	0.517531	0.132294	0.676137	ALL PASS
2-Bit Pair Distribution	0.287093	0.459664	0.765886	ALL PASS

2.4 Spectral and Structural Analysis

This category produced the only failure for Java SecureRandom. The Spectral FFT (Periodicity) test failed with p=0.008976, indicating detection of periodic components in the bitstream. BBRES-RNG and Math.random() both passed. The three remaining spectral tests passed for all generators.

Test	BBRES-RNG	SecureRandom	Math.random()	Status
Spectral FFT (Periodicity)	0.699220	**0.008976** ⚠	0.145508	**1 FAIL**
Compression Ratio (zlib)	0.218963	0.218963	0.218963	ALL PASS
Binary Derivative (1st order)	0.670910	0.376209	0.521536	ALL PASS
Turning Point Test	0.579502	0.849709	0.038991	ALL PASS

2.5 Entropy Analysis

All three generators achieved Shannon entropy above 99.998% of the theoretical 8-bit maximum. Transition rates were near-ideal for all generators.

Metric	BBRES-RNG	SecureRandom	Math.random()	Theoretical Max
Shannon Entropy (8-bit)	7.999860	7.999864	7.999878	8.000000
Min-Entropy (8-bit)	7.944862	7.947469	7.946686	8.000000
Transition Rate	0.500064	0.500134	0.499903	0.500000

2.6 Adversarial and Predictability Tests

All three generators passed all five adversarial tests and the SHA-256 CSPRNG wrapper validation. Machine learning attack accuracies hovered around 50%, confirming unpredictability.

Test	BBRES-RNG	SecureRandom	Math.random()	Status

Frequency Prediction (w=8)	p=0.524, Acc=50.20%	p=0.680, Acc=50.20%	p=0.708, Acc=50.20%	ALL PASS
ML Attack - LogReg (w=16)	p=0.976, Acc=49.01%	p=0.374, Acc=50.16%	p=0.500, Acc=50.00%	ALL PASS
ML Attack - GBT (w=16)	p=0.878, Acc=49.35%	p=0.776, Acc=49.58%	p=0.606, Acc=49.85%	ALL PASS
ML Attack - MLP (w=16)	p=0.174, Acc=50.52%	p=0.720, Acc=49.68%	p=0.903, Acc=49.28%	ALL PASS
Pattern Repetition (w=59)	p=1.000	p=1.000	p=1.000	ALL PASS
SHA-256 CSPRNG Wrapper	p=1.000, Acc=53.38%	p=1.000, Acc=53.07%	p=1.000, Acc=53.18%	ALL PASS

2.7 Integer Distribution Tests

This category produced the only failure for Java Math.random(): the Anderson-Darling test failed with p=0.000277 (A2*=1.6704), indicating a subtle departure from uniform distribution in the integer tail. All other distribution tests passed for all generators. BBRES-RNG and SecureRandom passed all eight tests.

Test	BBRES-RNG	SecureRandom	Math.random()	Status
Chi-Square Uniformity	0.475159	0.779776	0.865575	ALL PASS
Mean (Z-test)	0.398873	0.756969	0.305685	ALL PASS
Variance (Chi-sq)	0.842551	0.311325	0.307893	ALL PASS
Binned Goodness-of-Fit	0.756309	0.358056	0.788826	ALL PASS
Birthday Spacing	1.000000	1.000000	1.000000	ALL PASS
Coupon Collector	0.500000	0.500000	0.500000	ALL PASS
Collision Test	0.999851	0.999851	0.999851	ALL PASS
Anderson-Darling	0.357194	0.572926	**0.000277** ⚠️	**1 FAIL**

2.8 Integer Sequence Tests

All three generators passed all eight integer sequence tests, confirming no patterns, memory effects, or predictability in consecutive integer outputs.

Test	BBRES-RNG	SecureRandom	Math.random()	Status
Skewness / Kurtosis	0.937413	0.894683	0.858873	ALL PASS
Maximum-of-5	0.155701	0.023707	0.490623	ALL PASS
Runs Up / Down	0.994013	0.031462	0.812183	ALL PASS

Lag-1 Autocorrelation	0.418097	0.289195	0.214638	ALL PASS
Gap Test (KS)	0.332555	0.054087	0.996038	ALL PASS
Permutation (t=5)	0.703417	0.686551	0.076210	ALL PASS
Spearman Rank Correlation	0.417971	0.287360	0.213005	ALL PASS
Median Test	0.839534	0.775821	0.977283	ALL PASS

3. Failed Test Analysis

3.1 Java SecureRandom - Spectral FFT (Periodicity) Failure

Attribute	Detail
p-value	**0.008976 (below α = 0.01)**
Peaks below threshold	5,185,349 / 5,459,252
Interpretation	Periodic components were detected in the bitstream.
Impact	This is a narrow failure - the autocorrelation profile, entropy, and all other spectral sub-tests passed. The result may reflect a minor deterministic pattern introduced by SecureRandom's internal DRBG seeding cycle.
Crypto Impact	Low. SecureRandom passes all entropy, adversarial, and cryptographic wrapper tests. It remains suitable for cryptographic use.

3.2 Java Math.random() - Anderson-Darling Test Failure

Attribute	Detail
p-value	**0.000277 (well below α = 0.01)**
Statistic	A2* = 1.6704
Interpretation	The integer output exhibits a statistically significant departure from a uniform continuous distribution in the tail regions.
Root Cause	Math.random() uses a 48-bit Linear Congruential Generator. The LCG architecture produces subtle modular arithmetic patterns that affect the distribution of integer outputs in the tails.
Crypto Impact	High. The LCG architecture is deterministic and state-predictable. Math.random() is not suitable for cryptographic use regardless of statistical test outcomes.

4. Cross-Segment Consistency

Each bitstream was divided into 10 equal segments and independently tested for bit balance. All three generators demonstrated consistent quality across all segments.

Metric	BBRES-RNG	SecureRandom	Math.random()
KS Statistic	0.2097	0.3352	0.2103
KS p-value	0.6978	0.1671	0.6949
Cross-half correlation	r = 0.000099	r = -0.000621	r = 0.000111
Assessment	**Consistent**	**Consistent**	**Consistent**

5. Cryptographic Suitability Assessment

Criterion	BBRES-RNG	SecureRandom	Math.random()
Overall pass rate	**45/45 (100%)**	**44/45 (97.8%)**	**44/45 (97.8%)**
Shannon Entropy (% of max)	99.998%	99.998%	99.998%
Min-Entropy (% of max)	99.31%	99.34%	99.33%
ML Attack Resistance	~49-51% acc.	~49-50% acc.	~49-50% acc.
SHA-256 Wrapper	**PASS**	**PASS**	**PASS**
Architecture	Scheduler entropy	OS entropy + DRBG	Linear Congruential
Cryptographic Verdict	**SUITABLE**	**SUITABLE**	**NOT SUITABLE**

6. Usage Guidance

Use Case	BBRES-RNG	SecureRandom	Math.random()
Cryptographic key generation	**YES**	**YES**	**NO**
Session tokens / nonces	**YES**	**YES**	**NO**
Lottery / fairness-critical	**YES**	**YES**	**NO**
Monte Carlo simulations	**YES**	**YES**	**YES**
Game mechanics / UI randomness	**YES**	**YES**	**YES**
Statistical sampling	**YES**	**YES**	**YES**

7. Conclusions

BBRES-RNG

BBRES-RNG achieved a perfect 45/45 pass rate - the only generator to do so. It is the top-ranked generator in this analysis. The scheduler-based entropy harvesting architecture produced output that was statistically indistinguishable from true random across every evaluated dimension, including the Spectral FFT test where SecureRandom failed. BBRES-RNG is confirmed suitable for all applications, including cryptographic use.

Java SecureRandom

Java SecureRandom scored 44/45, failing only the Spectral FFT (Periodicity) test (p=0.008976). All other categories passed, including entropy, adversarial, and cryptographic wrapper tests. This isolated spectral anomaly does not undermine SecureRandom's suitability for cryptographic use. It remains a well-established, FIPS-aligned CSPRNG and a reliable standard for security-sensitive Java applications.

Java Math.random()

Java Math.random() scored 44/45, failing the Anderson-Darling test ($p=0.000277$, $A2^*=1.6704$). This failure arises from the inherent structural properties of its 48-bit LCG architecture, which produces subtle modular arithmetic patterns in the integer output domain. The LCG architecture is also deterministic once the seed is known, making it fundamentally unsuitable for cryptographic or high-assurance applications. It remains adequate for general-purpose, non-security-critical use cases such as simulations and UI randomness.

BBRES-RNG: #1 | SecureRandom: #2 | Math.random(): #3 (Not Crypto-Suitable)

Disclaimer: This report is an independent analysis based on submitted validation data for each RNG architecture. All tests used a significance level of α = 0.01 (99% confidence). Sample sizes: 10,918,505 bits and 100,000 integers per RNG.. BBRES-RNG developer: Mehul Singh.